FX Portfolio Lab:
From Portfolio Construction to FX Structuring

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Abstract

This working paper documents a research project that studies the foreign exchange (FX) market from three angles: the portfolio manager who allocates across currencies, the investor who hedges the currency risk of foreign assets, and the structurer who designs hedging products for corporate clients. We first build a valuation and backtesting engine whose correctness is verified against analytical results. We then construct G10 currency portfolios from carry, momentum and value signals and combine them with minimum-variance, mean-variance and risk-parity allocations, paying particular attention to estimation error and transaction costs. Next, we derive the optimal hedge ratio of a euro-based investor. Finally, we price and compare the structured products used to hedge FX exposures. Every design decision is recorded together with the alternatives considered and the reason for the choice (Appendix A).

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1 Introduction

The FX market is the largest financial market in the world, with an average daily turnover of about 7.5 trillion US dollars in April 2022 (Bank for International Settlements, 2022). Banks intervene in it in three main roles: they take and manage risk on behalf of investors, they help clients hedge their exposures, and they design structured solutions for corporate treasurers. This project reproduces, at a small scale and with public data, the analytical work behind each of these roles.

The project is organised around four research questions.

1. **Engine.** How can a currency portfolio be valued correctly, including forward positions, rebalancing and transaction costs? (Section 4)

2. **Currency portfolios.** Do the carry, momentum and value strategies documented in the literature (Lustig et al., 2011; Menkhoff et al., 2012a,b; Asness et al., 2013) survive realistic transaction costs, and which allocation method combines them best out of sample? (Section 5)

3. **Hedging.** Which fraction of the currency exposure of foreign equities and bonds should a euro-based investor hedge to minimise risk? (Campbell et al., 2010) (Section 6)

4. **Structuring.** How do forwards, options and structured products (collars, participating forwards, target redemption forwards) compare for a corporate hedger, in terms of cost and distribution of outcomes? (Section 7)

The analytical framework of the first two questions is that of portfolio construction theory: valuation of rebalanced portfolios, first two moments of a portfolio return, minimum-variance and mean-variance optimisation (Markowitz, 1952; Milhau, 2026). A recurring theme is that the inputs of these optimisations are estimated, and that naive mean-variance portfolios are dominated out of sample by simple rules (DeMiguel et al., 2009).

Methodological principles. Three principles are applied throughout. (i) *No look-ahead bias*: every quantity used to form a portfolio at date t is computed from data available at t. (ii) *Verification*: every piece of code is checked against an analytical result or a published number before it is used (Appendix B). (iii) *Traceability*: every choice that could change the results is recorded in the decision log (Appendix A).

2 Framework and notation

2.1 Portfolio valuation

A portfolio invested in N constituents holds an amount $\theta_{t,i}$ in constituent i at date t. Its value and weights are

$$V_t = \sum_{i=1}^{N} \theta_{t,i}, \qquad w_{t,i} = \frac{\theta_{t,i}}{V_t}. \tag{1}$$

Between two rebalancing dates the portfolio is buy-and-hold: the number of units of each constituent is constant, so each amount grows with the constituent's own return,

$$\theta_{[t+1]^-,i} = \theta_{t,i}\,(1 + r_{t,t+1,i}), \tag{2}$$

and the portfolio return is the average of the constituents' returns weighted by the *initial* weights,

$$r_{t,t+1,p} = \sum_{i=1}^{N} w_{t,i}\, r_{t,t+1,i} = \boldsymbol{w}_t' \boldsymbol{r}_{t,t+1}. \tag{3}$$

On a rebalancing date the amounts are reset by the allocation rule; for a fixed-mix rule with target weights w_i, $\theta_{t+1,i} = w_i V_{t+1}$. The amount traded, $\sum_i |\theta_{t+1,i} - \theta_{[t+1]^-,i}|$, determines the transaction costs.

2.2 Moments and optimal portfolios

With $\boldsymbol{\mu}$ the vector of expected returns and $\boldsymbol{\Sigma}$ the covariance matrix of the constituents,

$$\mu_p = \boldsymbol{w}'\boldsymbol{\mu}, \qquad \sigma_p^2 = \boldsymbol{w}'\boldsymbol{\Sigma}\boldsymbol{w}. \tag{4}$$

The global minimum variance (GMV) portfolio and the maximum return-to-risk (MRR) portfolio are

$$\boldsymbol{w}_{\text{gmv}} = \frac{\boldsymbol{\Sigma}^{-1}\mathbf{1}}{\mathbf{1}'\boldsymbol{\Sigma}^{-1}\mathbf{1}}, \qquad \boldsymbol{w}_{\text{mrr}} = \frac{\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}}{\mathbf{1}'\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}}, \tag{5}$$

and, by the two-fund separation theorem, every efficient portfolio is a combination of the two (Milhau, 2026).

2.3 Currency excess returns

Let S_t be the spot exchange rate, in US dollars per unit of foreign currency, and i_t, i_t^* the domestic (USD) and foreign interest rates over $(t, t+1)$. A US investor who borrows one dollar at i_t and invests it in a foreign deposit earns the excess return

$$rx_{t+1} = (1 + i_t^*)\frac{S_{t+1}}{S_t} - (1 + i_t) \approx \underbrace{(i_t^* - i_t)}_{\text{carry}} + \underbrace{\frac{S_{t+1} - S_t}{S_t}}_{\text{spot change}}. \tag{6}$$

The same exposure is obtained with a forward contract. Under covered interest parity (CIP) the forward rate is $F_t = S_t(1 + i_t)/(1 + i_t^*)$, and a long forward position on $1/F_t$ units of foreign currency yields

$$\frac{S_{t+1} - F_t}{F_t} = \frac{rx_{t+1}}{1 + i_t}. \tag{7}$$

The approximation in (6) shows the economic content of a currency position: the investor earns the interest rate differential and bears the risk of a depreciation of the foreign currency. Uncovered interest parity predicts that the two terms offset on average; the empirical failure of this prediction is the source of the carry trade premium (Lustig et al., 2011). Since 2008, CIP no longer holds exactly (Du et al., 2018); the consequences for our data are discussed in Section 3.

3 Data

All series are downloaded from FRED, the database of the Federal Reserve Bank of St. Louis, by the script `scripts/download_data.py` (Decision D7). Table 1 lists them.

Currency	Spot series	Spot available	Rate series	Rate available
USD	–	–	`IR3TIB01USM156N`	1964-06 – 2026-09
EUR	`DEXUSEU`	1999-01 – 2026-10	`IR3TIB01EZM156N`	1994-01 – 2026-01
GBP	`DEXUSUK`	1971-01 – 2026-10	`IR3TIB01GBM156N`	1957-01 – 2026-01
AUD	`DEXUSAL`	1971-01 – 2026-10	`IR3TIB01AUM156N`	1968-01 – 2026-08
NZD	`DEXUSNZ`	1971-01 – 2026-10	`IR3TIB01NZM156N`	1973-12 – 2026-09
JPY	`DEXJPUS`	1971-01 – 2026-10	`IR3TIB01JPM156N`	2002-04 – 2026-08
CHF	`DEXSZUS`	1971-01 – 2026-10	`IR3TIB01CHM156N`	1999-07 – 2026-09
CAD	`DEXCAUS`	1971-01 – 2026-10	`IR3TIB01CAM156N`	1956-01 – 2026-09
NOK	`DEXNOUS`	1971-01 – 2026-10	`IR3TIB01NOM156N`	1979-01 – 2026-09
SEK	`DEXSDUS`	1971-01 – 2026-10	`IR3TIB01SEM156N`	1982-01 – 2026-09

Table 1: Data series (downloaded on 19 October 2026). Spot: Federal Reserve H.10, daily noon buying rates in New York. Rates: OECD Main Economic Indicators, 3-month interbank rates, monthly averages, % per year.

Quotation convention. FRED quotes the euro, the pound and the Australian and New Zealand dollars in US dollars per unit of foreign currency, and the other currencies in foreign currency per US dollar. All spot rates are converted to US dollars per unit of foreign currency, so that an increase of S_t is always an appreciation of the foreign currency, consistently with equation (6).

Sample. The sample starts in January 1999, with the introduction of the euro. A currency enters the panel as soon as its spot rate and interest rate are both available (Decision D9): the Swiss franc in August 1999 and the yen in May 2002, all other currencies in February 1999. The euro and sterling rates published by the OECD stop in January 2026, so these two currencies leave the panel after February 2026. A balanced panel (May 2002 – February 2026) is kept as a robustness check.

3.1 Descriptive statistics

Table 2 reports the monthly excess returns of a long position in each currency against the US dollar, computed as described in Section 4.2. The row EW is the equally weighted portfolio of the available currencies, rebalanced monthly; its return is the cross-sectional average of the excess returns. In the literature it is called the *dollar factor*, since it is short the US dollar against all other currencies (Lustig et al., 2011).

Three facts emerge from Table 2 and Figure 1.

1. **Volatility.** Annualised volatilities lie between 8% and 13%, below those of an equity index (15–20%) but far from negligible: a currency position is a risky asset in its own right. The EW portfolio has a volatility of 8.1%, below that of every currency, as predicted by diversification.

	First month	Last month	Months	Mean (%/yr)	Carry (%/yr)	Vol (%/yr)	Sharpe	Worst month (%)	Max DD (%)
EUR	1999-02	2026-02	324	-0.07	-0.58	9.21	-0.01	-9.78	42.41
GBP	1999-02	2026-02	324	0.20	0.50	8.50	0.02	-8.98	43.64
AUD	1999-02	2026-09	331	2.48	1.50	11.69	0.21	-16.53	38.78
NZD	1999-02	2026-09	331	2.75	1.81	12.58	0.22	-12.72	33.98
JPY	2002-05	2026-09	292	-2.06	-1.69	9.26	-0.22	-8.16	62.43
CHF	1999-08	2026-09	325	0.79	-1.80	9.71	0.08	-11.18	32.22
CAD	1999-02	2026-09	331	0.42	-0.11	8.24	0.05	-12.90	35.49
NOK	1999-02	2026-09	331	0.50	0.93	11.39	0.04	-11.94	53.62
SEK	1999-02	2026-09	331	-1.11	-0.66	10.86	-0.10	-10.62	53.15
EW	1999-02	2026-09	331	0.35	–	8.10	0.04	-8.89	39.16

Table 2: Monthly currency excess returns against the US dollar, February 1999 – September 2026. Mean, carry and volatility are annualised ($\times 12$ and $\times\sqrt{12}$). Carry is the average forward discount $S_t/F_t - 1$, the excess return earned if the spot rate does not move. Sharpe is the annualised mean divided by the annualised volatility (the returns are already in excess of the US rate). Max DD is the maximum drawdown. Generated by `scripts/describe_returns.py`.

2. **Carry and returns line up.** The currencies with the highest carry (AUD, NZD) have the highest average excess returns, and the yen, the currency with the lowest carry, has the lowest. This is the pattern exploited by the carry trade (Section 5). The Swiss franc is an exception: its carry is negative, yet its excess return is positive, because it appreciated against the dollar; this is consistent with its status as a safe-haven currency. Figure 1 also shows the asymmetry of carry returns: the cumulative value of the Australian dollar position fell by 32% between June 2008 and January 2009, more than ten years of its average excess return (2.5% a year) lost in seven months, which is the crash risk documented by Brunnermeier et al. (2009).

3. **Average returns are imprecise.** The t-statistic of a mean return is approximately the annualised Sharpe ratio times the square root of the number of years. With Sharpe ratios below 0.25 and about 27 years of data, no individual mean is significantly different from zero at the 5% level ($0.22 \times \sqrt{27.5} \approx 1.15 < 1.96$). Expected returns are therefore much harder to estimate than volatilities, which is the practical argument for the minimum-variance portfolio (Milhau, 2026).

3.2 Limitations

- **Forward rates** are not freely available; they are implied by CIP (equation 7). Since 2008, CIP deviations of the order of a few tens of basis points per year exist (Du et al., 2018).
- **Interest rates** are monthly averages of interbank rates, not the rate observed on the day a position is opened.
- **Maturity mismatch**: 3-month rates are used to price 1-month forwards, because 1-month rates are not freely available for all currencies. The error on the carry is of the order of the slope of the money-market curve between 1 and 3 months.
- **Euro and sterling rates** stop in January 2026 in the OECD data; they could be extended with the ECB and Bank of England series.
- **Transaction costs**: bid–ask spreads are not in the data and must be modelled [TODO: reference and figures].

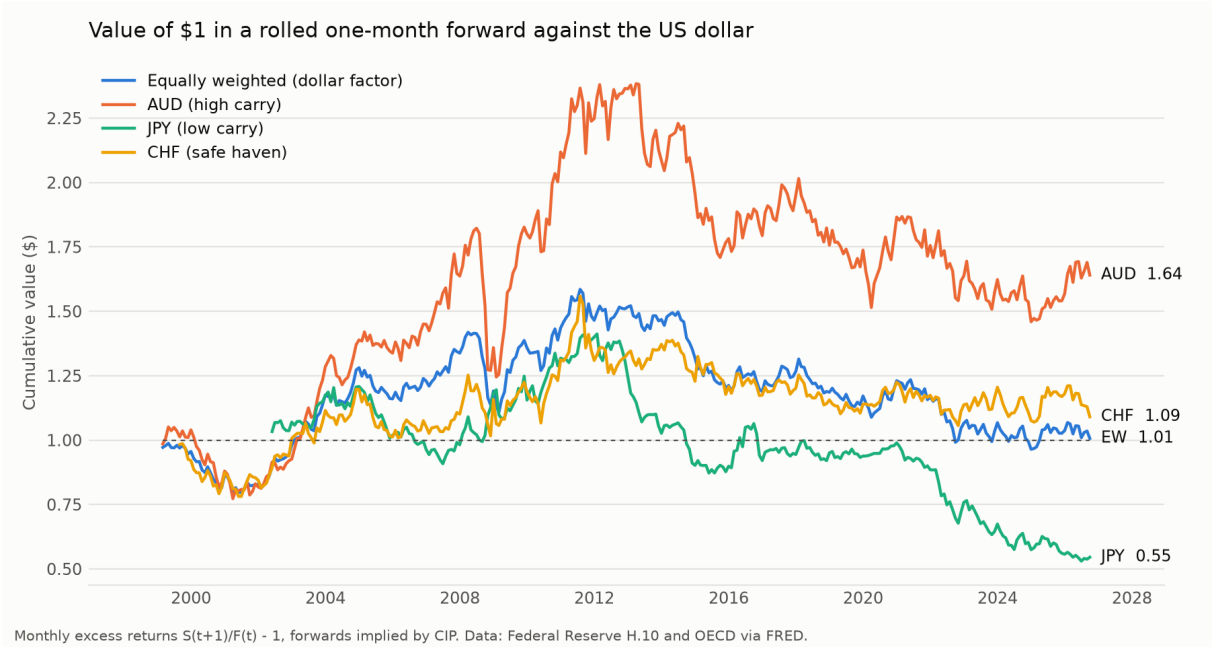


Figure 1: Cumulative value of $1 invested in a one-month forward rolled every month, for the equally weighted portfolio and three currencies with contrasting carry. Generated by `scripts/make_figures.py`.

4 Valuation and backtesting engine

4.1 Portfolio valuation

The engine values a portfolio by propagating the dollar amounts held in each constituent, following the general method of Milhau (2026): amounts are updated with equation (2) between rebalancing dates and reset by the allocation rule on rebalancing dates. Working with amounts rather than weights has two advantages. First, it handles any rebalancing frequency with the same code (daily rebalancing and buy-and-hold are the two limiting cases). Second, it gives the amount traded at each date directly, from which transaction costs are computed (Decision D4).

Verification. The engine reproduces the worked examples of the course: the valuation of an equally weighted portfolio of five US stocks between 2 and 5 January 2015 (final value $978.21, $6.37 traded to return to equal weights), the drift of weights in a buy-and-hold portfolio, and the compounding of returns. These examples are encoded as automated tests (Appendix B).

4.2 Monthly currency excess returns

Sampling. The month-end spot rate is the last available daily rate of the month (Decision D8); if the data stop before the last business day of the final month, that month is dropped, so that no mid-month rate is treated as a month-end rate. The OECD interest rate of month t is an average over the month, so it is dated at the end of month t, when it is fully known.

Forward rate and excess return. At the end of month t, the one-month forward rate implied by CIP is

$$F_t = S_t \frac{1 + i_t/12}{1 + i_t^*/12}, \tag{8}$$

and the excess return of a long forward position held over month $t+1$ is (Decision D11)

$$rx_{t+1} = \frac{S_{t+1}}{F_t} - 1. \tag{9}$$

The *carry* of the position, known at t, is $S_t/F_t - 1 \approx (i_t^* - i_t)/12$: the return earned if the spot rate does not move.

Timing. Equation (9) uses only S_t, i_t and i_t^*, observed at the end of month t, to open the position; the outcome S_{t+1} is observed one month later. A test checks that changing the interest rates at $t+1$ or the spot rates after $t+1$ leaves rx_{t+1} unchanged.

[TODO: Phase 1: forward positions in the backtester, transaction costs, performance and risk measures.]

5 Currency portfolios

[TODO: Phase 2.]

- Signals: carry (interest rate differential), momentum (past excess returns), value (deviation from purchasing power parity).
- Portfolio formation: sorting into long-short portfolios versus optimisation.
- Allocation across strategies: equal weights, GMV, mean-variance, risk parity; covariance estimation with shrinkage (Ledoit and Wolf, 2004).
- Crash risk of the carry trade (Brunnermeier et al., 2009).

6 Optimal currency hedging for a euro-based investor

[TODO: Phase 3.]

The minimum-variance hedge ratio is a direct application of the two-asset minimum-variance weight $w_2^* = (\sigma_1^2 - \sigma_{12})/\sigma_{2/1}^2$ (Milhau, 2026), where the two assets are the unhedged foreign asset and the currency forward. The empirical question is whether some currencies, such as the US dollar, act as natural hedges for equity risk (Campbell et al., 2010).

7 FX structuring

[TODO: Phase 4.]

- Option pricing: Garman and Kohlhagen (1983); volatility smile from at-the-money, risk-reversal and butterfly quotes.
- Products: forward, vanilla option, collar, participating forward, target redemption forward (Monte Carlo).
- Comparison for a corporate hedger: upfront cost, distribution of the hedged rate, worst-case outcomes.

8 Conclusion and limitations

[TODO: Written at the end of the project.]

Known limitations at this stage:

- Forward rates are implied by covered interest parity rather than observed.
- Transaction costs are modelled, not observed.

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A Decision log

Each entry records a choice that could affect the results or the usability of the project, the alternatives that were considered and the reason for the choice. Entries are never deleted: a decision that is reversed is superseded by a new entry that refers to it.

D1 – Language of the project

Date 18 October 2026

Decision The paper, the code and its comments are written in English.

Alternatives French.

Rationale English is the working language of trading floors and of open-source code, and the project is meant to be shown to recruiters, including outside France.

D2 – Code organised as an installable Python package

Date 18 October 2026

Decision The code lives in a package (`src/fxlab`) installed in a virtual environment; notebooks only import and use it.

Alternatives (a) Code written directly in Jupyter notebooks. (b) Loose scripts.

Rationale Code in notebooks cannot be tested or reused, and is easily run out of order. A package separates the library (tested, reused by every phase) from the analysis (notebooks). The `src` layout guarantees that the tests run against the installed package, not against stray files.

D3 – Course examples used as automated tests

Date 18 October 2026

Decision Every worked example of the course that the code can reproduce is encoded as a test (`pytest`), with a tolerance that reflects the rounding of the published inputs.

Alternatives Checking results by eye in notebooks.

Rationale The course examples have known answers, so they are an independent specification of what the code must do. Automated tests are rerun after every change, which prevents an improvement in one place from silently breaking another.

D4 – Valuation by propagation of amounts

Date 18 October 2026

Decision Portfolios are valued by propagating the amount held in each constituent and resetting it on rebalancing dates (course, section 1.8).

Alternatives Compounding the portfolio return $\boldsymbol{w}'\boldsymbol{r}$ with fixed weights.

Rationale Compounding with fixed weights is only valid when the portfolio is rebalanced at every date. Propagating amounts handles any rebalancing frequency and gives the amount traded, which is needed for transaction costs.

D5 – Free, official data sources

Date 18 October 2026

Decision Exchange rates and interest rates are taken from official publishers (Federal Reserve, OECD, ECB) through FRED and the ECB data portal; forward rates are implied by CIP.

Alternatives (a) Bloomberg or Refinitiv: no access. (b) Yahoo Finance: unofficial interface, no documentation of the series, known data errors.

Rationale Official sources are documented, stable and allow anyone to reproduce the results. The cost is that forward rates are not observed; this is acknowledged as a limitation (Section 3).

D6 – Working paper in LaTeX, versioned with the code

Date 18 October 2026

Decision The documentation is a LaTeX article stored in the same repository as the code.

Alternatives (a) A word processor document. (b) Notebooks only.

Rationale LaTeX is the standard for scientific documents with equations. Keeping it in the repository ties each version of the text to the version of the code that produced its results.

D7 – Data downloaded by a script from the FRED CSV interface

Date 19 October 2026

Decision A script downloads each series from its public FRED address and caches the raw file in `data/raw/`; the files are read and cleaned by the `fxlab.data` module. Spot rates are converted to US dollars per unit of foreign currency.

Alternatives (a) Manual download from the website. (b) The FRED API, which requires a personal key. (c) A third-party package such as `pandas-datareader`.

Rationale A script makes the dataset reproducible with one command and can refresh it at any time. The public CSV interface needs no key and only the Python standard library, so there is no dependency that could break. A single quotation convention avoids sign errors on half of the currencies.

D8 – Monthly frequency, month-end sampling

Date 19 October 2026

Decision Currency strategies are formed and evaluated monthly, using the last available spot rate of each month.

Alternatives (a) Daily frequency. (b) Weekly frequency. (c) Monthly averages of spot rates.

Rationale Interest rates are only available monthly, so a higher frequency would reuse the same rate for a whole month. Monthly rebalancing is the standard of the currency literature (Lustig et al., 2011; Menkhoff et al., 2012a), which makes the results comparable, and it keeps turnover and transaction costs low. Month-end rates, not averages, are used because a position is opened and closed at a given rate, not at an average.

D9 – Unbalanced panel from January 1999

Date 19 October 2026

Decision The sample starts in January 1999; each currency enters as soon as its spot rate and interest rate are both available. A balanced panel (2002–2026) is used as a robustness check.

Alternatives (a) Balanced panel from May 2002 only. (b) Start in the 1980s without the euro.

Rationale Starting with the euro gives a homogeneous G10 universe. An unbalanced panel uses three more years of data without inventing any, as in Lustig et al. (2011); the robustness check verifies that the conclusions do not depend on this choice.

D10 – US dollar as base currency for the currency portfolios

Date 19 October 2026

Decision The excess returns of Phase 2 are measured from the point of view of a US investor; the euro-based investor is the subject of Phase 3.

Alternatives Euro as base currency from the start.

Rationale The data are quoted against the US dollar and almost all of the literature uses a US investor, so the results can be compared with published ones. Excess returns of long-short portfolios do not depend much on the base currency, since the base currency cancels out of a long-short position.

D11 – Excess return defined as the simple return of a one-month forward

Date 19 October 2026

Decision $rx_{t+1} = S_{t+1}/F_t - 1$ with F_t implied by CIP and the monthly rate equal to the annual rate divided by 12 (equations 8–9).

Alternatives (a) Log excess return $\ln S_{t+1} - \ln F_t$, common in the literature. (b) Deposit version (6).

Rationale The portfolio formula $r_p = \boldsymbol{w}'\boldsymbol{r}$ of the course holds for simple returns, not for log returns, so simple returns can be aggregated across currencies without approximation. The forward is the instrument actually traded by an FX desk for this exposure, and it requires no capital beyond margin. The deposit version differs only by the factor $1 + i_t$ (equation 7).

D12 – Continuous integration and MIT licence

Date 19 October 2026

Decision The test suite runs automatically on GitHub Actions at every push, on Python 3.11 and 3.13; the code is published under the MIT licence.

Alternatives (a) Running the tests locally only. (b) No licence, or a restrictive one.

Rationale Tests run on a clean machine catch errors that a local environment hides (a missing dependency, a file that only exists locally), and the badge shows any reader that the suite passes. Without a licence, published code cannot legally be reused; MIT is the most common licence for demonstration projects and imposes only attribution.

D13 – Compiled paper versioned in the repository

Date 19 October 2026

Decision The compiled PDF is copied to `paper/fx-portfolio-lab.pdf` by `make` and committed with each update of the paper; LaTeX build files remain ignored.

Alternatives (a) PDF attached to GitHub releases only. (b) PDF compiled by the continuous integration. (c) Sources only.

Rationale A reader who is not going to compile LaTeX must be able to open the paper in one click from the README. Committing the PDF is the simplest way to guarantee it and keeps each version of the PDF next to the code that produced it. Compiling in the CI would require installing a TeX distribution on every run, which is slow for a document that changes only a few times per phase.

Open decisions

- Transaction cost model (bid–ask spreads on spot and forwards).
- Extension of the euro and sterling interest rates after January 2026 (ECB and Bank of England series).

B Software architecture and verification

Module	Content
`fxlab.portfolio`	Returns, compounding, valuation of buy-and-hold and rebalanced portfolios
`fxlab.data`	Download and loading of FRED series, quotation convention
`fxlab.fx_returns`	Month-end sampling, CIP forwards, excess returns, carry

Table 3: Modules of the `fxlab` package.

Test	Source	Result checked
`test_table1_buy_and_hold`	Course, Table 1	$V_1 = \$978.21$, return -2.18%
`test_table1_rebalancing_...`	Course, section 1.4	\$6.37 traded
`test_midterm_q1_amount_update`	Midterm 2024–25, Q1	\$36.29
`test_midterm_q4_weight_drift`	Midterm 2024–25, Q4	weight 74.9%
`test_midterm_q7_compounding`	Midterm 2024–25, Q7	+0.8%
`test_single_rebalance_...`	Course, section 1.8	equals buy-and-hold
`test_spot_converted_...`	FRED conventions	JPY per USD inverted
`test_missing_values_...`	FRED format	missing values are NaN
`test_rates_in_decimal`	FRED format	5% read as 0.05
`test_forward_by_cip`	Eq. (8)	forward computed by hand
`test_excess_return_by_hand`	Eq. (9)	excess return computed by hand
`test_equal_rates_...`	Eq. (9)	no carry: rx = spot return
`test_carry_...`	Section 4.2	carry $\approx (i^* - i)/12$
`test_no_look_ahead`	Timing rule	rx_{t+1} unaffected by later data
`test_month_end_spot_...`	Sampling rule	last available day used
`test_incomplete_last_month_...`	Sampling rule	partial month dropped
`test_rates_dated_...`	Timing rule	rate dated at month end

Table 4: Verification tests.